# MH1201 Linear Algebra II

Midterm Exam (Solutions)

AY 2021/2022, Semester 2

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**Overview**

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- Vector spaces and subspaces in matrix spaces: subspace test, closure properties, counterexamples.
- Linear transformations on polynomial spaces: evaluation maps, matrix representations with respect to bases, rank–nullity, kernel and range.
- Solution structure: each question is solved using multiple independent methods (standard/explanatory first, then a compact alternative; a third method is included when natural).
- Notation: null($T$) and range($T$) denote the kernel and image of $T$, respectively.

# Contents

# Question 1 (10 Marks)

## Problem

For a matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R}),$$

and $i \in \{1, 2, \dots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \cdots + a_{in}.$$

In other words, $r_i(A)$ is the sum of entries in the ith row of $A$.

(a) Consider the set $\mathcal{U} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 1 \text{ for all } i = 1, 2, \dots, n\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(b) Consider the set $\mathcal{V} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) \geq 0 \text{ for all } i = 1, 2, \dots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(c) Consider the set $\mathcal{W} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 0 \text{ for some } i = 1, 2, \dots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

## Solution

### Method 1: Subspace test by axioms

Recall: a nonempty subset $S$ of a vector space is a subspace iff it contains $\mathbf{0}$ and is closed under addition and scalar multiplication.

(a) **(Failure: $0 \notin \mathcal{U}$).** The zero matrix 0 satisfies $r_i(0) = 0$ for all $i$, so $0 \notin \mathcal{U}$ (since $\mathcal{U}$ requires $r_i(A) = 1$ for all $i$). Hence $\mathcal{U}$ is not a subspace.

$$\boxed{\mathcal{U} \text{ is not a subspace of } \mathcal{M}_{n\times n}(\mathbb{R}).}$$

(b) **(Failure: not closed under scalar multiplication).** Take $A \in \mathcal{V}$ with at least one row sum $r_i(A) > 0$, e.g. let $A$ be the matrix whose first row is $(1, 0, \dots, 0)$ and all other entries are 0. Then $A \in \mathcal{V}$, but $(-1)A$ has $r_1((-1)A) = -1 < 0$, so $(-1)A \notin \mathcal{V}$. Thus $\mathcal{V}$ is not a subspace.

$$\boxed{\mathcal{V} \text{ is not a subspace of } \mathcal{M}_{n\times n}(\mathbb{R}).}$$

(c) If $n = 1$, then $\mathcal{W} = \{[0]\}$, which *is* a subspace of $\mathcal{M}_{1\times 1}(\mathbb{R}) \cong \mathbb{R}$.

Assume now that $n \geq 2$. **(Failure: not closed under addition).** Define diagonal matrices

$$A = \operatorname{diag}(0, 1, 1, \dots, 1), \qquad B = \operatorname{diag}(1, 0, 1, \dots, 1).$$

Then $r_1(A) = 0$, so $A \in \mathcal{W}$; and $r_2(B) = 0$, so $B \in \mathcal{W}$. However,

$$A + B = \operatorname{diag}(1, 1, 2, \dots, 2),$$

so every row sum of $A + B$ is positive (hence nonzero), meaning $A + B \notin \mathcal{W}$. Therefore $\mathcal{W}$ is not a subspace for $n \geq 2$.

$$\boxed{\mathcal{W} \text{ is a subspace iff } n = 1;\ \text{for } n \geq 2,\ \mathcal{W} \text{ is not a subspace.}}$$

**Method 2: Linear-map viewpoint**

Define the linear map

$$R : \mathcal{M}_{n\times n}(\mathbb{R}) \to \mathbb{R}^n, \qquad R(A) = (r_1(A), \dots, r_n(A)).$$

Each $r_i$ is linear in $A$, hence $R$ is linear. Also recall: if $S \le \mathbb{R}^n$ is a subspace, then $R^{-1}(S) \le \mathcal{M}_{n\times n}(\mathbb{R})$.

(a) $\mathcal{U} = R^{-1}(\mathbf{1})$ where $\mathbf{1} = (1, \dots, 1) \neq \mathbf{0}$. Since $\{\mathbf{1}\}$ is not a subspace of $\mathbb{R}^n$ (it does not contain $\mathbf{0}$), it follows that $\mathcal{U}$ cannot be a subspace.

$$\boxed{\mathcal{U} \text{ is not a subspace.}}$$

(b) $\mathcal{V} = R^{-1}(\mathbb{R}^n_{\ge 0})$. The set $\mathbb{R}^n_{\ge 0}$ is not a subspace of $\mathbb{R}^n$ (not closed under multiplication by $-1$), hence $\mathcal{V}$ is not a subspace.

$$\boxed{\mathcal{V} \text{ is not a subspace.}}$$

(c) Let $H_i = \{y \in \mathbb{R}^n : y_i = 0\}$, a subspace of $\mathbb{R}^n$. Then

$$\mathcal{W} = R^{-1}\Big(\bigcup_{i=1}^{n} H_i\Big) = \bigcup_{i=1}^{n} R^{-1}(H_i),$$

a union of subspaces. For $n \ge 2$, $\bigcup_{i=1}^n H_i$ is not a subspace of $\mathbb{R}^n$ (e.g. $e_1 \in H_2$ and $e_2 \in H_1$ but $e_1 + e_2 \notin \bigcup_i H_i$), hence $\mathcal{W}$ is not a subspace for $n \ge 2$. For $n = 1$, $\bigcup_i H_i = \{0\}$, so $\mathcal{W}$ is a subspace.

$$\boxed{\mathcal{W} \text{ is a subspace iff } n = 1; \text{ otherwise not.}}$$

**Method 3: Affine + Union theorem shortcuts**

We use two standard results:

**(T1) Affine vs linear subspace.** If $L : V \to W$ is linear and $w \in W$, then $L^{-1}(w)$ is either empty or an *affine* subspace of $V$; it is a *linear* subspace iff $w = 0$ (equivalently iff $0 \in L^{-1}(w)$).

**(T2) Union of subspaces.** If $U, V$ are subspaces of a vector space, then $U \cup V$ is a subspace iff $U \subseteq V$ or $V \subseteq U$.

With $R$ as in Method 2, apply these:

(a) $\mathcal{U} = R^{-1}(\mathbf{1})$ with $\mathbf{1} \neq 0$. By (T1), $\mathcal{U}$ is an affine translate of $\ker(R)$, not a linear subspace. Equivalently, $0 \notin \mathcal{U}$.

$$\boxed{\mathcal{U} \text{ is not a subspace.}}$$

(b) $\mathcal{V} = R^{-1}(\mathbb{R}^n_{\geq 0})$ where $\mathbb{R}^n_{\geq 0}$ is a cone but not a subspace (since $y \in \mathbb{R}^n_{\geq 0} \Rightarrow -y \notin \mathbb{R}^n_{\geq 0}$ unless $y = 0$). Hence $\mathcal{V}$ cannot be a subspace.

$$\boxed{\mathcal{V} \text{ is not a subspace.}}$$

(c) For each $i$, set $U_i := R^{-1}(H_i)$ where $H_i = \{y \in \mathbb{R}^n : y_i = 0\}$. Since $H_i$ is a subspace and $R$ is linear, each $U_i$ is a subspace. Moreover,

$$\mathcal{W} = \bigcup_{i=1}^{n} U_i.$$

If $n = 1$, then $\mathcal{W} = U_1 = \ker(R)$ is a subspace.

Assume $n \geq 2$. Consider $U_1$ and $U_2$. Neither contains the other: indeed, choose $A$ with $r_1(A) = 0$ and $r_2(A) = 1$ (so $A \in U_1 \setminus U_2$), and choose $B$ with $r_2(B) = 0$ and $r_1(B) = 1$ (so $B \in U_2 \setminus U_1$). Hence $U_1 \cup U_2$ is not a subspace by (T2), and therefore $\mathcal{W}$ (which contains $U_1 \cup U_2$) is not a subspace for $n \geq 2$.

$$\boxed{\mathcal{W} \text{ is a subspace iff } n = 1; \text{ for } n \geq 2, \ \mathcal{W} \text{ is not a subspace.}}$$

## Mark Scheme (10 marks)

- **(a) (3 marks)**
  - (1) States subspace requirement $0 \in \mathcal{U}$ (or equivalent subspace test).
  - (1) Computes/notes $r_i(0) = 0$ for all $i$.
  - (1) Concludes $0 \notin \mathcal{U} \Rightarrow$ not a subspace.
- **(b) (3 marks)**
  - (1) Identifies closure under scalar multiplication as required.
  - (1) Provides valid $A \in \mathcal{V}$ with some $r_i(A) > 0$ (any correct example).
  - (1) Shows $(-1)A \notin \mathcal{V}$ and concludes not a subspace.
- **(c) (4 marks)**
  - (1) Correctly handles $n = 1$: $\mathcal{W} = \{[0]\}$ hence subspace.
  - (1) For $n \geq 2$, identifies a necessary subspace property to test (closure under addition, or union-of-subspaces criterion).
  - (1) Produces $A, B \in \mathcal{W}$ with zero row-sum in different rows (any correct construction).
  - (1) Shows $A + B \notin \mathcal{W}$ (or applies theorem: union of incomparable subspaces is not a subspace), concluding not a subspace for $n \geq 2$.

# Question 2 (20 Marks)

## Problem

Let $T : P_4(\mathbb{R}) \to \mathbb{R}^3$ be the map defined by

$$T(f) = (f(0), f(1), f(2)).$$

(a) Show that $T$ is a linear map.

(b) Consider the following basis $\mathcal{B} = \{1, x, x^2, x^3, x^2(x-1)(x-2)\}$ for $P_4(\mathbb{R})$ (you need not show that $\mathcal{B}$ is a basis). Let $\mathcal{A} = \{e_1, e_2, e_3\}$ be the standard basis for $\mathbb{R}^3$. Write down the matrix representation of $T$ with respect to $\mathcal{B}$ and $\mathcal{A}$.

(c) Determine the dimensions of null$(T)$ and range$(T)$. In your solution, you must provide a basis for either null$(T)$ or range$(T)$. You need not provide bases for both null$(T)$ and range$(T)$.

(d) Determine whether the following properties of $T$ is true. You are to explain your answer.

  (i) Is $T$ injective?

  (ii) Is $T$ surjective?

  (iii) Is $T$ invertible?

## Solution

### Method 1: Direct evaluation + factor theorem

(a) Let $f, g \in P_4(\mathbb{R})$ and let $\lambda \in \mathbb{R}$. Then

$$T(f + \lambda g) = \big((f + \lambda g)(0), (f + \lambda g)(1), (f + \lambda g)(2)\big).$$

By linearity of polynomial evaluation,

$$(f + \lambda g)(a) = f(a) + \lambda g(a) \qquad (a = 0, 1, 2).$$

Hence

$$T(f + \lambda g) = \big(f(0) + \lambda g(0),\ f(1) + \lambda g(1),\ f(2) + \lambda g(2)\big) = T(f) + \lambda T(g).$$

Therefore $T$ is linear.

$$\boxed{T \text{ is a linear map.}}$$

(b) To obtain the matrix of $T$ with respect to domain basis $\mathcal{B}$ and codomain basis $\mathcal{A}$, we compute $T$ on each element of $\mathcal{B}$ and use the resulting coordinate vectors in $\mathcal{A}$ as the columns.

Since $\mathcal{A} = \{e_1, e_2, e_3\}$ is the standard basis of $\mathbb{R}^3$, the coordinate vector of a vector in $\mathbb{R}^3$ is just the vector itself.

Now:

$$\begin{aligned} T(1) &= (1,1,1), \\ T(x) &= (0,1,2), \\ T(x^2) &= (0,1,4), \\ T(x^3) &= (0,1,8), \end{aligned}$$

and

$$T\big(x^2(x-1)(x-2)\big) = (0,0,0),$$

because $x^2(x-1)(x-2)$ vanishes at $x = 0, 1, 2$.

Therefore

$$\boxed{[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}}$$

(c) First determine the null space.

By definition,

$$\text{null}(T) = \{f \in P_4(\mathbb{R}) : f(0) = f(1) = f(2) = 0\}.$$

Thus every $f \in \text{null}(T)$ has roots $0, 1, 2$. By the factor theorem,

$$x, \quad x-1, \quad x-2$$

all divide $f(x)$, so

$$f(x) = x(x-1)(x-2)q(x)$$

for some polynomial $q(x)$.

Since $\deg f \leq 4$, we must have $\deg q \leq 1$. Hence

$$q(x) = ax + b \qquad (a, b \in \mathbb{R}),$$

so

$$f(x) = x(x-1)(x-2)(ax+b).$$

Therefore

$$\text{null}(T) = \text{span}\{x(x-1)(x-2),\, x^2(x-1)(x-2)\}.$$

So a basis for $\text{null}(T)$ is

$$\{x(x-1)(x-2),\, x^2(x-1)(x-2)\},$$

and hence

$$\dim \text{null}(T) = 2.$$

Now use rank–nullity:

$$\dim P_4(\mathbb{R}) = \dim \text{null}(T) + \dim \text{range}(T).$$

Since $\dim P_4(\mathbb{R}) = 5$, we get

$$5 = 2 + \dim \operatorname{range}(T),$$

so

$$\dim \operatorname{range}(T) = 3.$$

Thus

$$\boxed{\dim \operatorname{null}(T) = 2, \qquad \dim \operatorname{range}(T) = 3.}$$

(d) (i) $T$ is injective iff $\operatorname{null}(T) = \{0\}$. But

$$\dim \operatorname{null}(T) = 2 > 0,$$

so $\operatorname{null}(T) \neq \{0\}$. Therefore $T$ is not injective.

$$\boxed{T \text{ is not injective.}}$$

(ii) Since $\operatorname{range}(T)$ is a subspace of $\mathbb{R}^3$ and

$$\dim \operatorname{range}(T) = 3 = \dim \mathbb{R}^3,$$

it follows that

$$\operatorname{range}(T) = \mathbb{R}^3.$$

Therefore $T$ is surjective.

$$\boxed{T \text{ is surjective.}}$$

(iii) A linear map is invertible iff it is bijective, i.e. both injective and surjective. Since $T$ is surjective but not injective, it is not invertible.

$$\boxed{T \text{ is not invertible.}}$$

**Method 2: Matrix method + rank–nullity**

(a) Let $f, g \in P_4(\mathbb{R})$ and let $\lambda \in \mathbb{R}$. Then

$$T(f+\lambda g) = \big((f+\lambda g)(0), (f+\lambda g)(1), (f+\lambda g)(2)\big).$$

Since polynomial evaluation is linear at each point,

$$(f+\lambda g)(0) = f(0)+\lambda g(0), \qquad (f+\lambda g)(1) = f(1)+\lambda g(1), \qquad (f+\lambda g)(2) = f(2)+\lambda g(2).$$

Therefore

$$T(f+\lambda g) = \big(f(0)+\lambda g(0),\ f(1)+\lambda g(1),\ f(2)+\lambda g(2)\big) = T(f)+\lambda T(g).$$

Hence $T$ is linear.

$$\boxed{T \text{ is linear.}}$$

(b) Let

$$\mathcal{B} = \{1, x, x^2, x^3, x^2(x-1)(x-2)\}, \qquad \mathcal{A} = \{e_1, e_2, e_3\}.$$

To find the matrix of $T$ with respect to $\mathcal{B}$ and $\mathcal{A}$, we compute the image of each basis vector in $\mathcal{B}$ and place its coordinate vector relative to $\mathcal{A}$ as a column.

Now,

$$T(1) = (1,1,1),$$
$$T(x) = (0,1,2),$$
$$T(x^2) = (0,1,4),$$
$$T(x^3) = (0,1,8),$$

and

$$T\big(x^2(x-1)(x-2)\big) = (0,0,0),$$

because

$$x^2(x-1)(x-2)$$

vanishes at $x = 0, 1, 2$.

Therefore

$$\boxed{[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}}$$

(c) Using the matrix found in part (b),

$$[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}.$$

Row-reducing:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 2 & 4 & 8 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 6 & 0 \end{bmatrix}.$$

There are 3 pivot rows, so

$$\operatorname{rank}(T) = 3.$$

Hence

$$\dim \operatorname{range}(T) = 3.$$

Since

$$\dim P_4(\mathbb{R}) = 5,$$

rank–nullity gives

$$\dim \operatorname{null}(T) = 5 - 3 = 2.$$

To provide a basis for range$(T)$, take the pivot columns of the original matrix. These are the first three columns, so a basis for range$(T)$ is

$$\{(1,1,1), (0,1,2), (0,1,4)\}.$$

Thus

$$\operatorname{range}(T) = \operatorname{span}\{(1,1,1), (0,1,2), (0,1,4)\},$$

and

$$\boxed{\dim(\operatorname{null}(T)) = 2, \qquad \dim(\operatorname{range}(T)) = 3.}$$

(d) (i) A linear map is injective iff its null space is trivial. Since

$$\dim \operatorname{null}(T) = 2 > 0,$$

we have

$$\operatorname{null}(T) \neq \{0\}.$$

Therefore $T$ is not injective.

$$\boxed{T \text{ is not injective.}}$$

(ii) A subspace of $\mathbb{R}^3$ of dimension 3 must be all of $\mathbb{R}^3$. Since

$$\dim \operatorname{range}(T) = 3,$$

it follows that

$$\operatorname{range}(T) = \mathbb{R}^3.$$

Therefore $T$ is surjective.

$$\boxed{T \text{ is surjective.}}$$

(iii) A linear map is invertible iff it is both injective and surjective. Since $T$ is surjective but not injective, $T$ is not invertible.

$$\boxed{T \text{ is not invertible.}}$$

**Method 3: Interpolation for the range + factor theorem for the kernel**

This method determines the range by interpolation and the null space by roots.

(a) Let $f, g \in P_4(\mathbb{R})$ and let $\lambda \in \mathbb{R}$. Then

$$T(f + \lambda g) = \big((f + \lambda g)(0), (f + \lambda g)(1), (f + \lambda g)(2)\big).$$

Using linearity of evaluation at each point,

$$(f+\lambda g)(0) = f(0)+\lambda g(0), \qquad (f+\lambda g)(1) = f(1)+\lambda g(1), \qquad (f+\lambda g)(2) = f(2)+\lambda g(2).$$

Hence

$$T(f + \lambda g) = \big(f(0) + \lambda g(0),\ f(1) + \lambda g(1),\ f(2) + \lambda g(2)\big) = T(f) + \lambda T(g).$$

Therefore $T$ is linear.

$$\boxed{T \text{ is linear.}}$$

(b) Let

$$\mathcal{B} = \{1, x, x^2, x^3, x^2(x-1)(x-2)\}, \qquad \mathcal{A} = \{e_1, e_2, e_3\}.$$

We compute $T$ on each vector in $\mathcal{B}$:

$$\begin{aligned} T(1) &= (1,1,1), \\ T(x) &= (0,1,2), \\ T(x^2) &= (0,1,4), \\ T(x^3) &= (0,1,8), \\ T\big(x^2(x-1)(x-2)\big) &= (0,0,0). \end{aligned}$$

Thus the matrix representation of $T$ with respect to $\mathcal{B}$ and $\mathcal{A}$ is

$$[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}.$$

Therefore

$$\boxed{[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{bmatrix}}$$

(c) Let $(y_0, y_1, y_2) \in \mathbb{R}^3$ be arbitrary. Define the Lagrange basis polynomials for the nodes $0, 1, 2$ by

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}, \\ \ell_1(x) &= \frac{x(x-2)}{(1-0)(1-2)} = -x(x-2), \\ \ell_2(x) &= \frac{x(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}. \end{aligned}$$

These satisfy

$$\ell_0(0) = 1,\ \ell_0(1) = 0,\ \ell_0(2) = 0,$$

$$\ell_1(0) = 0,\ \ell_1(1) = 1,\ \ell_1(2) = 0,$$
$$\ell_2(0) = 0,\ \ell_2(1) = 0,\ \ell_2(2) = 1.$$

Now define

$$p(x) = y_0\ell_0(x) + y_1\ell_1(x) + y_2\ell_2(x).$$

Then

$$p(0) = y_0, \qquad p(1) = y_1, \qquad p(2) = y_2.$$

Also $\deg p \le 2$, so $p \in P_4(\mathbb{R})$. Therefore

$$T(p) = (y_0, y_1, y_2).$$

Since $(y_0, y_1, y_2)$ was arbitrary, it follows that

$$\operatorname{range}(T) = \mathbb{R}^3.$$

Hence

$$\dim \operatorname{range}(T) = 3.$$

A basis for $\operatorname{range}(T)$ is therefore

$$\{e_1, e_2, e_3\}.$$

Now determine the null space. By definition,

$$\operatorname{null}(T) = \{f \in P_4(\mathbb{R}) : f(0) = f(1) = f(2) = 0\}.$$

So every $f \in \operatorname{null}(T)$ has roots $0, 1, 2$. By the factor theorem,

$$f(x) = x(x-1)(x-2)q(x)$$

for some polynomial $q(x)$. Since $\deg f \le 4$, we must have $\deg q \le 1$, so

$$q(x) = ax + b \qquad (a, b \in \mathbb{R}).$$

Hence

$$f(x) = a\,x^2(x-1)(x-2) + b\,x(x-1)(x-2),$$

which shows that

$$\operatorname{null}(T) = \operatorname{span}\{x(x-1)(x-2),\ x^2(x-1)(x-2)\}.$$

Therefore

$$\dim \operatorname{null}(T) = 2.$$

Thus

$$\boxed{\dim \operatorname{null}(T) = 2, \qquad \dim \operatorname{range}(T) = 3.}$$

(d) (i) A linear map is injective iff its null space is $\{0\}$. Since

$$\dim \operatorname{null}(T) = 2 > 0,$$

the null space is nontrivial. Therefore $T$ is not injective.

$$\boxed{T \text{ is not injective.}}$$

(ii) Since

$$\operatorname{range}(T) = \mathbb{R}^3,$$

the map $T$ is surjective.

$$\boxed{T \text{ is surjective.}}$$

(iii) A linear map is invertible iff it is bijective. Since $T$ is surjective but not injective, it is not invertible.

$$\boxed{T \text{ is not invertible.}}$$

## Mark Scheme (20 marks)

- **(a) Linearity (4 marks)**
  - (1) States the correct linearity condition.
  - (2) Correctly computes $T(f+\lambda g)$ componentwise.
  - (1) Concludes that $T(f+\lambda g) = T(f) + \lambda T(g)$, hence $T$ is linear.
- **(b) Matrix representation (5 marks)**
  - (1) Recognises that columns are the images of the basis vectors in $\mathcal{B}$.
  - (1) Correctly computes $T(1) = (1, 1, 1)$.
  - (1) Correctly computes $T(x) = (0, 1, 2)$ and $T(x^2) = (0, 1, 4)$.
  - (1) Correctly computes $T(x^3) = (0, 1, 8)$.
  - (1) Correctly computes $T(x^2(x-1)(x-2)) = (0, 0, 0)$ and writes the full matrix.
- **(c)** $\text{null}(T)$ **and** $\text{range}(T)$ **(6 marks)**
  - (2) Correctly characterises $\text{null}(T)$ as polynomials vanishing at $0, 1, 2$, and uses factor theorem or an equivalent argument.
  - (1) Gives a correct basis for $\text{null}(T)$ *or* a correct basis for $\text{range}(T)$.
  - (1) Correctly obtains $\dim \text{null}(T) = 2$.
  - (2) Correctly obtains $\dim \text{range}(T) = 3$ by rank–nullity, rank, or surjectivity.
- **(d) Injective / surjective / invertible (5 marks)**
  - (1) Correctly states that $T$ is not injective.
  - (1) Correctly justifies non-injectivity using $\text{null}(T) \neq \{0\}$ or $\dim \text{null}(T) = 2$.
  - (1) Correctly states that $T$ is surjective.
  - (1) Correctly justifies surjectivity using $\dim \text{range}(T) = 3$ or $\text{range}(T) = \mathbb{R}^3$.
  - (1) Correctly concludes that $T$ is not invertible, with justification.